

OPEN PHYSICS RESEARCH COLLECTIVE

Decentralized Research Cluster: Matter & Materials

OFFICIAL VERDICT: RFC #1

Subject: The Absence of Free Electrons in Ideal Insulators

1. Collective Conclusion

Per the operational mandate of the Founding Charter, the OPRC has conducted a three-phase verification of electronic states in insulating crystals.

Final Verdict: Ideal insulators explicitly do not contain free electrons. The conduction band remains statistically vacant under standard conditions, and all valence electrons are spatially localized to atomic sites.

2. Computational Proofs

The following data was generated dynamically by the OPRC Digital Lab:

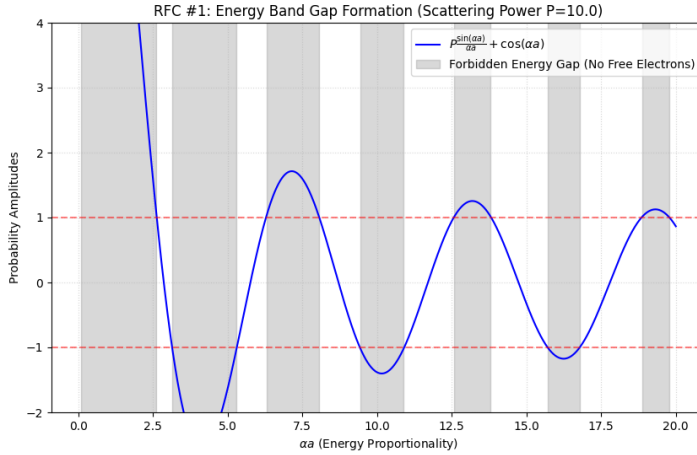


Figure 1: Phase 1: Proof of Forbidden Energy Zones where $|\cos(ka)| > 1$.

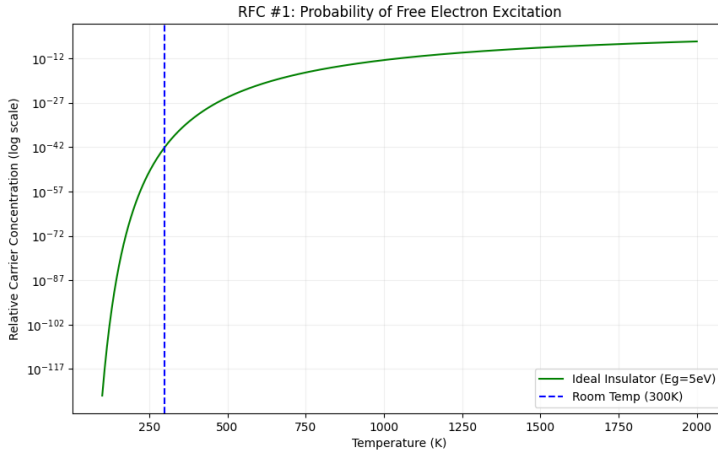


Figure 2: Phase 2: Statistical probability of conduction band occupancy ($n_c \approx 10^{-87}$).

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3. Certification

This verdict is issued under the Open Source Mandate. All mathematical derivations are verified reproducible via the OPRC containerization protocol.

Authenticated by OPRC Genesis Branch

Timestamp: May 3, 2026

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